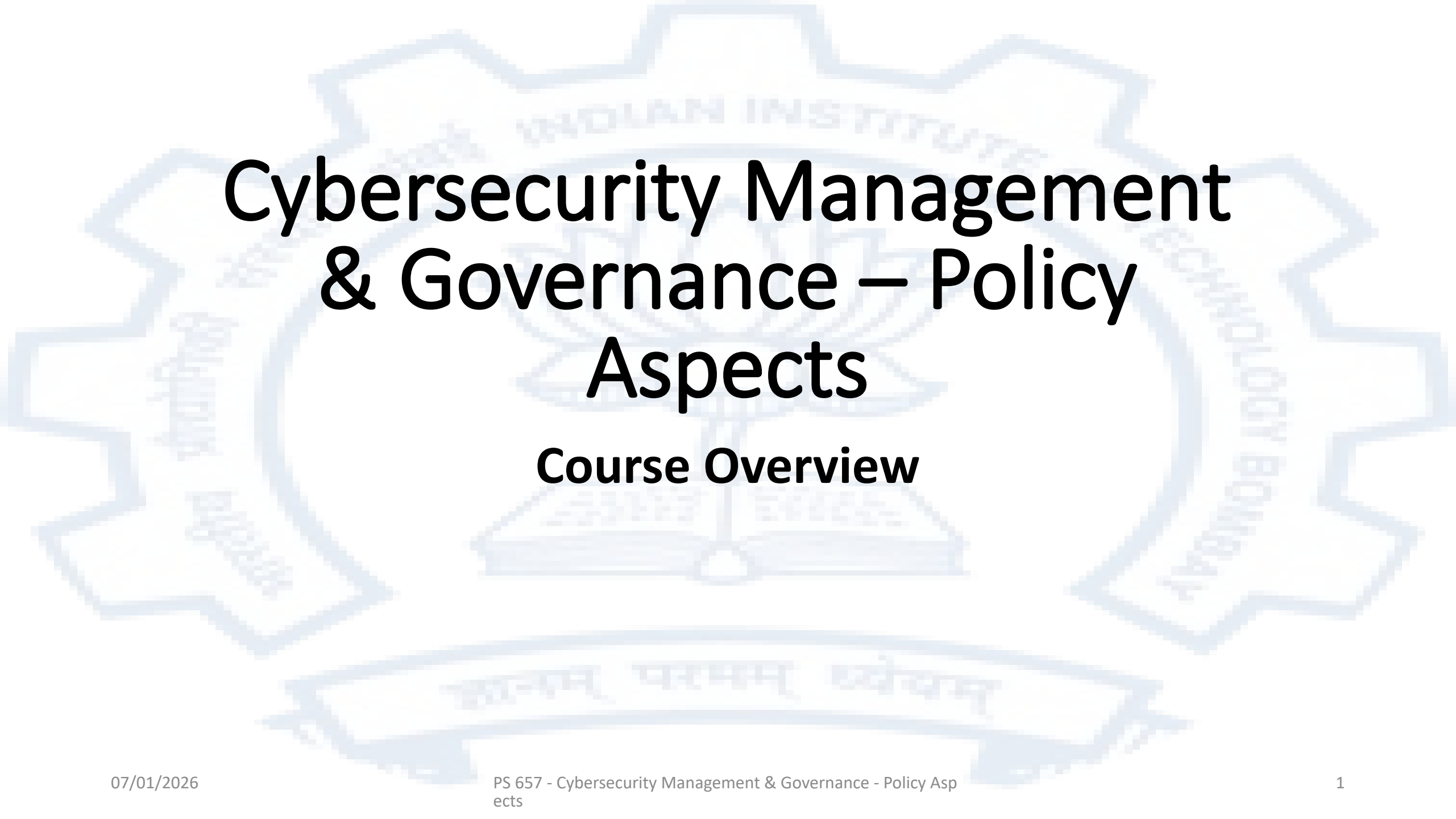




Cybersecurity Management & Governance – Policy Aspects

Course Overview

Information Security

- The need and ability to control information and its dissemination has been recognized from the earliest of human civilizations
- Powerful people or groups have used information to exercise control over large numbers of people
- The earliest of societies have used various methods to ensure that sensitive information does not fall into the hands of potential adversaries or into the hands of elements who may use the information to the disadvantage of others
- Typical examples of use of sensitive information have been to pass secret communications for military purposes, guard knowledge that the guardians feel that only a special group is entitled to know and give the possessor an advantage in negotiation or in some other way

Historical methods of information security

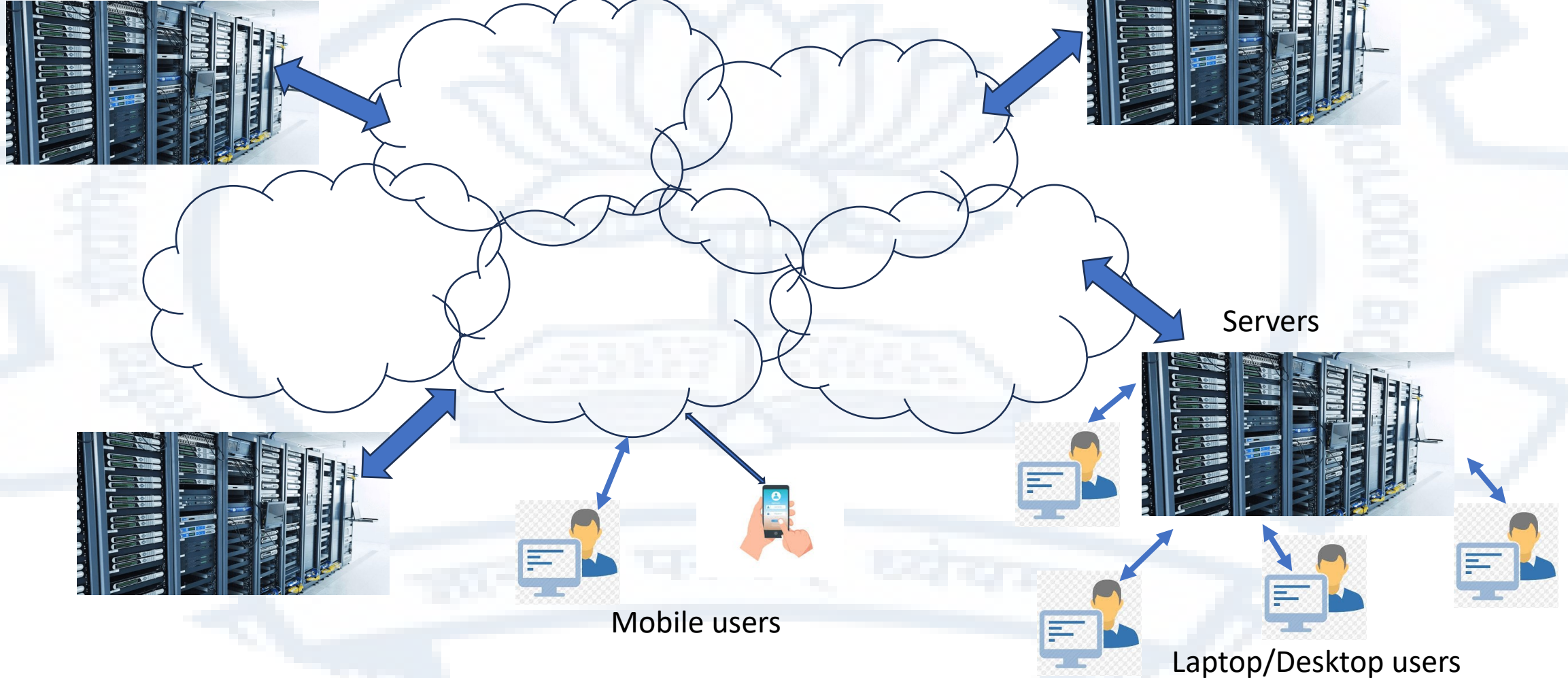
- Ensure that unauthorized people do not have access to the information
 - Use of secure physical containers to store information
 - Use of other clever concealment methods for information
 - Social norms that denied the training to decipher and use information
- Transforming the information into another form that is not decipherable without special knowledge
 - Codes of varying degrees of sophistication were known to and in use by early Babylonian, Egyptian, Indian and Greek civilizations
 - The modern versions of this technique “Cryptology” (Cryptography+Cryptanalysis) form the bedrock of information security today

Information Security becomes a well defined subject

- The development of concepts such as a computing machine and the early realizations of such machines allowed for much more sophistication in developing algorithms for encryption
- With the increasing deployment of computers and the increasing reliance on data stored in computers the subject of information security began to take on a more formal hue

Modern Information Systems

Networks of networks



Information security evolves to Cybersecurity

- With the advent of modern computers and networking, the number of ways to access or intercept data has increased exponentially
- Additionally, since these systems operate on behalf of software entities called “users”, we need techniques to associate the software “user” to real human beings
- Software systems are complex and are prone to having bugs which may enable unauthorized access or result in harmful malfunctioning
- We need to ensure that data is protected throughout its life and that only authorized people can use services and data

Cybersecurity

Cybersecurity may be defined as the practice of protecting data, systems, networks and software from malicious attacks or malfunctions. These attacks are usually aimed at accessing, changing, or destroying sensitive information or at gaining access to and using services without being authorized to.

An important aspect of cybersecurity is information security and many of the techniques of information security form the bedrock of many cybersecurity techniques

Course Structure

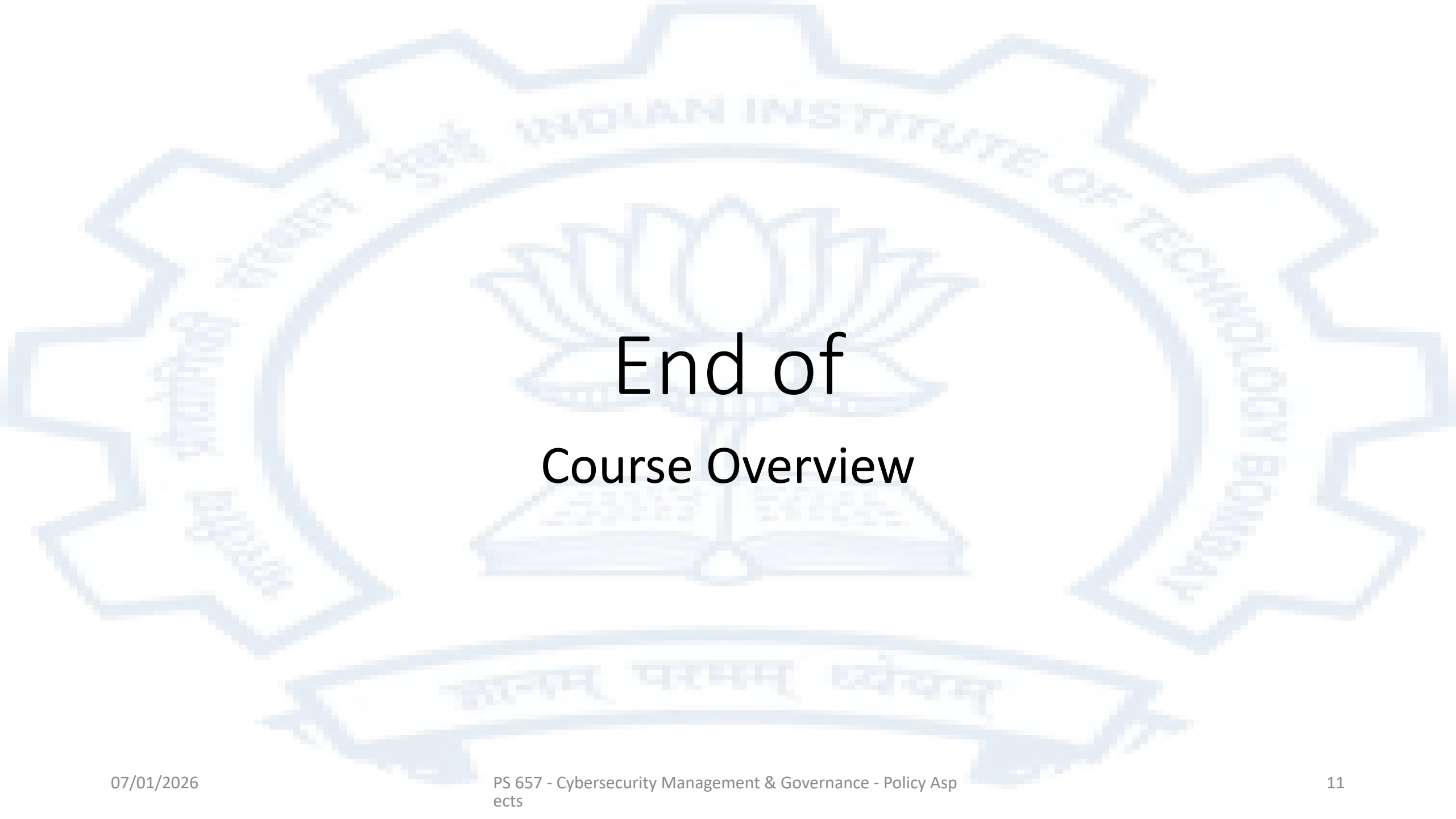
- Basic cybersecurity principles and techniques
- Governance and Management of cybersecurity
- Policy aspects of cybersecurity management and governance

Course Material

- Cybersecurity Body of Knowledge CyBOK v 1.1.0 – is available for download at https://www.cybok.org/media/downloads/CyBOK_v1.1.0.pdf
- Lecture slides – will be uploaded after each lecture to Moodle
- Other reading material that will be periodically uploaded to Moodle

Course Evaluation

- Attendance – 20%
 - 0 for less than 50% attendance
 - Linear extrapolation for 50+% to 100% attendance
- Internal evaluation – 30%
 - Could be group work and/or individual work
- End Semester Exam – 50%

The background of the slide features a large, light blue watermark of the Indian Institute of Technology Bombay (IIT Bombay) logo. The logo is circular with a gear-like outer border. Inside the circle, there is a lotus flower in the center, and the text "INDIAN INSTITUTE OF TECHNOLOGY BOMBAY" is written in a circular path around the lotus. At the bottom of the logo, there is a banner with the Sanskrit motto "वैद्यमान् परममन् व्योमन्".

End of Course Overview